# Trade Study: Camera Options

- **Conducted by:** Matt Potts (mp1011698@wcupa.edu)
- **Decision:** IMX219-160

**Problem:** *The autonomous vehicle currently lacks a vision-based perception system. To fulfill this capability gap for this milestone, this trade study will evaluate and select a vehicle camera architecture. The evaluation will compare the procurement of integrated, multi-sensor development kits against standalone cameras. The chosen solution must maximize sensor data fidelity and compatibility with the Nvidia Jetson, while minimizing software integration lead time, mechanical integration footprint and staying within the budget. This trade study will identify options within both, and then come to an overall conclusion as well as a back up conclusion dependent on the selection of car.*

| | | *IMX219-160* | *IMX219 NoIR* | *Innomaker* |
|---|---|---|---|---|
| **Criteria** | **Weight** | **Grade** | **Grade** | **Grade** |
| *Resolution* | .05 | 9 | 9 | 4 |
| *Frame Rate* | .05 | 8 | 8 | 4 |
| *Field of View* | .10 | 8 | 10 | 5 |
| *Connection type* | .10 | 10 | 10 | 5 |
| *Compatibility* | .12 | 10 | 10 | 7 |
| *Size & weight* | .05 | 10 | 9 | 5 |
| *Latency* | .08 | 10 | 10 | 7 |
| *Power Consumption* | .10 | 9 | 7 | 5 |
| *Focus type* | .05 | 8 | 8 | 9 |
| *Complexity* | .10 | 10 | 10 | 5 |

| | | | | |
|---|---|---|---|---|
| *Weather Resistance* | .05 | 1 | 1 | 1 |
| *Documentation* | .05 | 10 | 8 | 4 |
| *Cost* | .05 | 9 | 8 | 10 |
| **Average Score** | x | 8.6 | 8.3 | 5.4 |
| **Weighted Score** | x | 8.6 | 8.2 | 5.7 |

**Breakdown**

**1st Choice:** *WaveShare IXM219-160*
*Resolution: Delivers 8MP, which meets AV standard. Loses a point for frame rate drop at maximum resolution.*
*Frame Rate: Good support for resolutions provided. Struggles at max resolution however will most likely not be operating at max resolution.*
*Field of view: 160 degrees is second highest in this trade study.*
*Connection type: Native MIPI CSI-2 Ribbon Cable.*
*Compatibility: Officially supported on Jetson.*
*Latency: Ribbon connection provides 20-30ms latency.*
*Power Consumption:*
*Size and weight: 25x24mm, 15g is minimal burden.*
*Cost: Lowest of all candidates, also included in developer kit.*
*Focus type: Fixed focus is good, but could struggle if object is close*
*Integration: Plug and play, native drivers, plenty of documentation.*
*Weather resistance: bare module provides no protection. It would need a case.*

Based on these results, the WaveShare IXM219-160 is the best option as it narrowly beats out the Arducam IXM219 NoIR. Although the Arducam camera has slightly better features, the features selection does not beat out the Arducam for the increase in price. Because of this, the WaveShare IXM219-160 is the recommended choice.

**2nd Choice:** *Arducam IXM219 NoIR*
*Resolution: Delivers same 8MP sensor as Waveshare.*
*Frame Rate: Delivers same profiles as Waveshare.*
*Field of view: 175 degrees is highest FOV of all options.*
*Connection type: Native MIPI CSI-2 Ribbon Cable.*
*Compatibility: Officially supported on Jetson.*
*Latency: Ribbon connection provides 20-30ms latency.*
*Power Consumption: Peaks at 300mA @ 3.3V, highest of all options. Acceptable but higher on power budget.*
*Size and weight: 25x24mm, 20g. Slightly heavier.*

*Cost: $28-40 is could be double the Waveshare which would only be for slightly higher gains.*
*Focus type: Same focus type as Waveshare.*
*Integration: Plug and play, native drivers, plenty of documentation.*
*Weather resistance: bare module provides no protection. It would need a case.*

The runner up Arducam IXM219 NoIR model is a close second choice. It delivers similar specifications to the WaveShare camera but has some slightly better features (like IR) and also an increased FOV which allows for better object detection. However, the extra cost does not justify the features especially for this project, so therefore it would not be the recommended choice unless budget allows for some extra room to allow for better object detection specifications.

**3rd Choice:** *Innomaker* 720P USB 2.0 UVC Camera
*Resolution: Lowest resolution of all options.*
*Frame Rate: Doesn't offer higher options in framerate or resolution to work with.*
*Field of view: 120 degrees is lowest of all options. Limits object detection.*
*Connection type: USB will add latency and require more power draw.*
*Compatibility: Not officially supported and uses generic drivers. Will be recognized but more headroom to get working.*
*Latency: USB could be working at 50-100ms of latency. Not very important for simple tasks but in a safety critical environment it would under perform.*
*Power Consumption: ~200mA @ 5V. Requires more power draw than other options.*
*Size and weight: 32x32mm and 60g weight is higher than the others.*
*Cost: $12-18 is the lowest of the options, but has big compromises*
*Focus type: Fixed with a slightly higher range.  No autofocus variants.*
*Integration: WIll require more drivers and power management.*
*Weather resistance: bare module provides no protection. It would need a case.*

The Innomaker 720p camera is the worst overall option and therefore not recommended. With the worst specs overall it gives us very little room for improvement in the future over the previous ones as the others give higher resolution options for optimization of features in the future. Additionally, the USB option takes away from ports that could be used for other sensors and has higher power consumption options for only a few dollars saved.